

now much more comfortable relying on third parties for their primary computing resources. The use of cloud computing services reduces their overall computing costs, including the costs of maintaining and upgrading their own systems and of preventing security breaches.

Q What is the ultimate impact of open source in the present software landscape?

The software landscape has seen a mix of proprietary and open source software throughout this century. In some areas, notably data management, open source projects have offered innovative and powerful solutions that are more advanced than those from proprietary vendors. Customers are choosing these technologies as much for their features as for the availability of the source code, which they are not likely to modify. But many customers are not comfortable with the community-based support found in many open source projects and prefer to pay for commercial support from a company that can offer guaranteed levels of service quality as well as have the ability to make any needed code changes to address critical software bugs. Open source projects are not likely to drive out proprietary software, if only because so many organisations have huge existing investments in those products. But free and open source software is likely to continue to be the driving force for collaborative projects and new technologies from R&D groups in academia and industry.

Q What are the vital concerns raised by the software industry that are being addressed by the Center for Open Source Investigation (COSI) at Carnegie Mellon University, Silicon Valley? The Center for Open Source Investigation (COSI) has a longstanding primary focus on evaluation, adoption, and the use of open source software by companies, governments and other organisations. A second important area is education about open source software.

The OSSPAL project (originally the Business Readiness Rating) is intended to help people find high-quality open source projects, and the newer FLOSSBOK project is aimed at identifying the wide range of topics related to open source software, along with sources for more information on them. Both projects rely on financial donations to support their work, and on volunteers to move them forward.

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Success stories in the use of any open source product (not just software) encourage people to see if they might benefit from following the same idea.
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Q How do open source developments help academia enhance skills in young talent?

Well-written open source code provides students with good examples of how to write code in a particular programming language and style. Also, as students try to join existing projects, they can learn about the various approaches to collaboration and about the tools that are used to track issues, manage the parts of a project, build new versions, and release them to the community.

Overall, open source development helps students to gain insights into the technology and process of software development, as well as on open source licences. They get excited the first time they make a contribution that is accepted into the project code base, and that serves as a motivation for continuing to participate in open source communities.

Q What are the major obstacles programmers face when adapting to the open source culture?

Many open source projects are not very

welcoming of new members. Initial contributions to such projects are rejected with harsh criticism that discourages people from continuing participation. Some projects have developed a 'toxic culture' that discriminates against various types of people. Also, projects tend to be monolingual, for instance, only in English or Hindi, so a person who is not fluent in that language may find it difficult to join the project. Even in more welcoming projects, programmers must adjust to the possibility that their contribution may not be accepted by the project committers.

Q Is it difficult for big organisations to completely rely on FOSS?

Yes. There are some business areas such as enterprise resource planning and tax reporting in which proprietary solutions are more advanced than FOSS projects. Also, certain proprietary applications are the *de facto* standard for some common tasks. These applications include Adobe Photoshop, which has a dominant position for image manipulation and a vast community of customers who have been trained in its use. While there are indeed some FOSS alternatives to Photoshop available easily, they are widely regarded as not having the power offered by the proprietary solution.

In short, there is no reason for an organisation to rely completely on one type of solution. Organisations should instead be looking for the best solution for a specific need. It may be proprietary, open source, or a combination of the two. Similarly, the solution may run on the organisation's own systems, in the cloud, or as a hybrid of the two.

Q Do you think there are still security concerns related to adopting open source?

Yes, this is the case even though various studies show that open source is at least as secure as proprietary software, and major security breaches have often occurred in proprietary systems. The fear is not rational, but most organisations are